PROJECT REPORT

RETAIL INVENTORY ANALYTICS

Inventory Demand Analysis and Forecasting

A Data Analytics Project using Python, Pandas and Data Visualization

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Academic Year	2026

Submitted in partial fulfilment of the requirements for the academic project.

2. Student Declaration

We declare that this project report, titled Retail Inventory Analytics and Inventory Demand Analysis and Forecasting, documents the work undertaken as part of our academic project. The report describes the dataset, analytical workflow, visualizations, observations, limitations and possible improvements associated with the project.

The analysis and numerical results included in this report are generated from the dataset available in the project directory. Where a particular metric or field is unavailable in the dataset, it is identified as unavailable rather than being presented as a measured result.

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3. Acknowledgement

We express our sincere gratitude to our faculty members and institution for providing the opportunity to undertake this project. The project helped us understand how data analytics can be applied to a practical business problem involving retail inventory and product demand.

We also acknowledge the developers of the open-source tools used in this work, including Python, Pandas, NumPy and Matplotlib. These tools support data processing, numerical analysis and visualization, making it possible to transform tabular data into more understandable analytical results.

Finally, we acknowledge the project materials, dataset and existing analysis scripts maintained in the project repository. The charts included in this report are taken from the existing project files whenever the expected image files are available.

4. Abstract

Retail businesses must balance product availability with inventory cost. Insufficient stock can lead to missed sales, while excessive stock can increase storage requirements and tie up working capital. Historical demand data can help a business understand product movement and plan inventory more systematically.

This project examines retail inventory data using Python-based data analysis. The workflow includes loading the dataset, inspecting its structure, reviewing missing values and duplicates, examining product and category demand, studying demand trends, and interpreting existing forecast and visualization outputs.

The project uses Pandas for tabular data handling and Matplotlib-generated charts for communicating patterns. A seven-period moving average is useful for smoothing short-term fluctuations, while a simple historical-average forecast can provide a baseline against which more advanced forecasting methods may be evaluated.

The report summarizes the actual dataset structure and available chart outputs. The reliability of any forecast depends on data quality, the forecast horizon, seasonality, and comparison with observations not used to create the forecast.

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6. Introduction

Inventory management is an important activity in retail operations. A retailer must maintain enough stock to meet customer demand while avoiding unnecessary accumulation of products. Decisions about purchasing, replenishment and storage become more difficult when many products have different demand patterns.

Data analytics helps convert historical records into useful information. By summarizing sales, comparing product demand and observing changes over time, analysts can identify products that may need closer attention.

6.1 Project Overview

The project focuses on retail inventory analytics and demand forecasting. It uses a structured dataset and existing Python analysis outputs to examine recorded demand, category patterns, moving averages and forecast comparisons.

6.2 Business Context

A retail manager needs to know which products sell quickly, whether demand is changing, and whether inventory planning should be reviewed. Historical demand can support these decisions, although it cannot guarantee future demand.

6.3 Importance of the Project

- Supports a structured understanding of historical demand.
- Helps identify products and categories for further review.
- Demonstrates practical data cleaning and visualization.
- Introduces basic time-series analysis and forecasting.
- Connects analytical results with inventory decisions.

7. Problem Statement and Objectives

7.1 Problem Statement

Retailers may find it difficult to plan inventory using raw transaction records alone. Without an organized view of product demand and its movement over time, decisions about replenishment may rely heavily on intuition. This project uses historical data analysis to make demand patterns easier to examine.

7.2 Main Objective

The main objective is to analyze available retail inventory data, identify demand patterns, review forecasting outputs and present the results in a readable analytical report.

7.3 Specific Objectives

- Load and inspect the retail inventory dataset.
- Identify missing values, duplicates and data-type issues.
- Summarize demand at product and category levels where fields permit.
- Examine demand changes over time.
- Explain the role of a seven-period moving average.
- Review the existing forecast-versus-actual visualization.
- Translate supported findings into practical recommendations.
- Document limitations and possible project extensions.

7.4 Expected Outcome

The expected outcome is a reproducible analytical workflow supported by tables, charts and a report that distinguishes measured results from interpretations and recommendations.

8. Scope and Business Requirements

8.1 Project Scope

The project covers analysis of the available dataset and visualization files. It includes descriptive statistics, product or category comparisons when relevant fields exist, time-based analysis when a usable date column exists, and interpretation of the forecasting outputs already generated by the project.

8.2 Business Questions

- Which products have the highest recorded demand?
- Which categories contribute the most recorded revenue, if revenue data exists?
- How does demand change over the available time period?
- What information does a moving average provide?
- How do forecast values compare with actual observations?
- Which products or categories deserve closer monitoring?
- What data-quality issues could affect the analysis?

8.3 Functional Requirements

Requirement	Purpose
Dataset loading	Read the available cleaned data.
Data inspection	Review columns, missing values and duplicates.
Demand summaries	Summarize available product and category measures.
Visualization	Present patterns using existing project charts.
Forecast review	Interpret the forecast comparison where available.
Report generation	Create a readable PDF with findings and references.

8.4 Out of Scope

The report generator does not automatically connect to live store systems, update stock in real time, or guarantee future sales. Those capabilities would require additional data sources, application logic and operational integration.

9. Technology Stack

Technology	Role in the Project
Python	Main programming language for the analysis workflow.
Pandas	Reading, cleaning and summarizing tabular data.
NumPy	Numerical operations where required by analysis scripts.
Matplotlib	Creating visualizations from the data.
VS Code	Development environment used to maintain project files.
ReportLab	Generating the formatted PDF report.
CSV / Excel	Storing the dataset in tabular form.

9.1 Why Python?

Python provides a readable environment for data preparation, statistical summaries and visualization. Its libraries allow the analytical workflow to be automated and rerun when the dataset changes.

9.2 Why Pandas?

Pandas provides DataFrame structures for handling rows and columns, checking missing values, removing duplicates, converting data types and grouping observations for summaries.

9.3 Why Visualization?

Charts make comparisons and trends easier to communicate than long tables of numbers. Their interpretation must still be grounded in the labels, scales and data used to generate them.

9.4 Report Automation

ReportLab is used to create page layouts, headings, tables, footers and chart placements. This script can be rerun from the project folder to refresh the report after the dataset or chart outputs are updated.

10. Dataset Description and Data Dictionary

The report uses the dataset found by the script in the project directory. The cleaned CSV is preferred when available. The table below describes the actual dataset structure detected during execution.

Dataset Metric	Observed Value
Number of records	73,501
Number of columns	15
Duplicate rows currently present	195
Missing cells currently present	160
Distinct products	20
Distinct categories	6
Date range	01 Jan 2025 to 01 Jan 2027
Total recorded units	10,028,853.00
Average units per record	136.45
Average recorded stock	274.45

10.1 Available Fields

Column Name	Detected Data Type
Date	str
Store ID	str
Product ID	str
Category	str
Region	str
Inventory Level	float64
Units Sold	float64
Units Ordered	int64
Demand Forecast	float64
Price	float64
Discount	float64
Weather Condition	str
Holiday/Promotion	int64
Competitor Pricing	float64
Seasonality	str

10.2 Data Interpretation

The meaning of a field should be confirmed from the dataset documentation or the analysis code. Column names alone may not fully describe how a measure was calculated.

11. Project Methodology

The project follows a structured data analytics workflow. Each stage prepares information for the next stage and helps keep the results reproducible.

Stage	Activity	Purpose
1. Load	Read the available dataset.	Make records available for analysis.
2. Inspect	Review columns, types and sample records.	Understand the dataset structure.
3. Clean	Check missing values and duplicates.	Identify data-quality problems.
4. Prepare	Convert relevant fields to suitable types.	Enable valid calculations.
5. Analyze	Group and summarize available measures.	Identify demand and category patterns.
6. Forecast	Review the existing forecast workflow.	Estimate near-term demand where implemented.
7. Visualize	Use the generated chart outputs.	Communicate the observed patterns.
8. Report	Document results and limitations.	Support clear interpretation and review.

11.1 Reproducibility

The report generator reads data and charts from the local project folder. This keeps the output connected to the files maintained in VS Code. For reproducible results, the dataset and chart-generating scripts should be preserved alongside the report.

11.2 Analytical Principle

A result should be included as a finding only when it can be supported by the available data or project output. Business recommendations are interpretations and should not be confused with directly measured values.

12. Data Quality and Preprocessing

Data quality affects the reliability of analysis. Missing values can reduce the number of usable observations, duplicate rows can inflate totals, and incorrect data types can prevent valid time-based or numerical calculations.

12.1 Current Dataset Checks

Check	Observed Result
Rows in selected CSV	73,501
Columns in selected CSV	15
Duplicate rows currently present	195
Missing cells currently present	160

12.2 Recommended Cleaning Activities

- Confirm that date fields use a consistent date format.
- Convert demand, revenue and stock fields to numeric values.
- Inspect duplicate records before deciding whether to remove them.
- Check missing values in important fields.
- Investigate negative or implausible numerical values.
- Standardize category labels when they refer to the same group.
- Record the number of rows before and after cleaning.

12.3 Important Note

The statistics above describe the CSV selected by this report generator at runtime. They do not by themselves prove what cleaning operations were performed earlier. The project's preprocessing scripts should be consulted to document those operations accurately.

13. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is used to understand the distribution, structure and relationships within a dataset before drawing conclusions. In inventory analysis, EDA can help reveal which products move quickly, whether demand changes over time, and whether some categories behave differently from others.

13.1 Descriptive Summary

Dataset Metric	Observed Value
Number of records	73,501
Number of columns	15
Duplicate rows currently present	195
Missing cells currently present	160
Distinct products	20
Distinct categories	6
Date range	01 Jan 2025 to 01 Jan 2027
Total recorded units	10,028,853.00
Average units per record	136.45
Average recorded stock	274.45

13.2 Product-Level Analysis

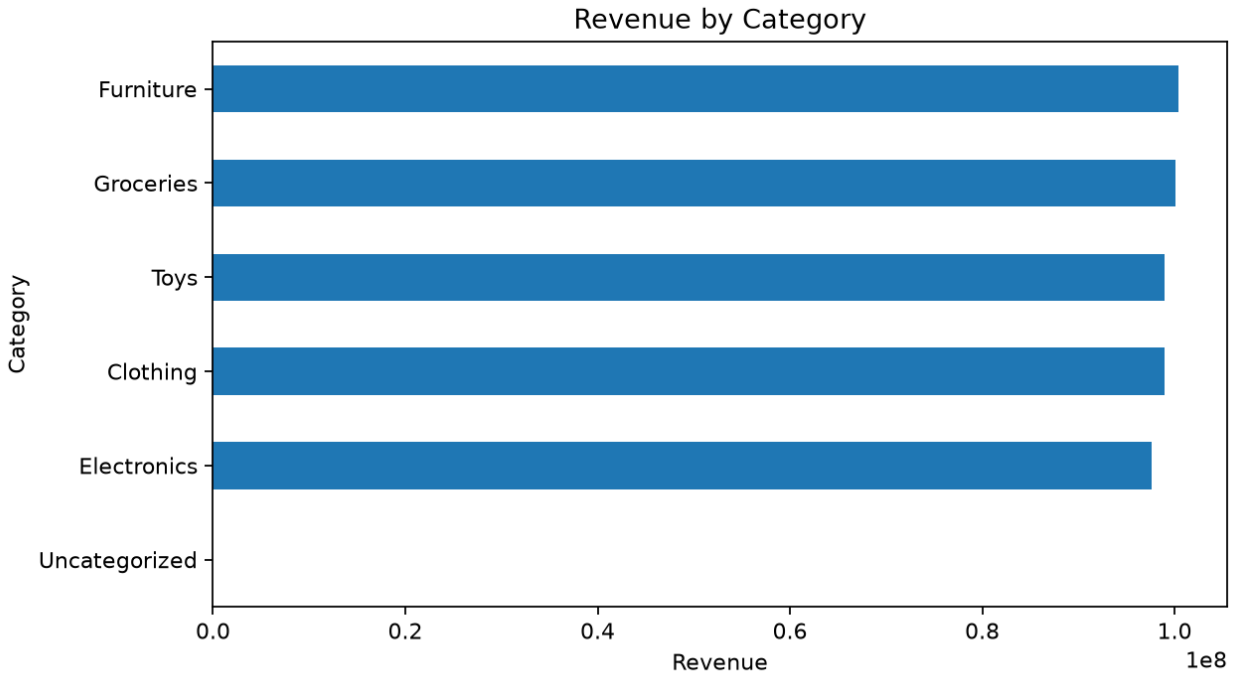
The following table summarizes total recorded units by product for the selected dataset. It is calculated from the actual CSV used by this script.

Product ID	Total Units Sold
P0016	511,891.00
P0015	511,377.00
P0014	510,122.00
P0020	509,998.00
P0005	506,490.00
P0013	504,228.00
P0009	503,376.00

13.3 Interpretation

Descriptive totals help identify relative demand, but they should be interpreted alongside the time period, stock availability and any missing records. A product with high recorded sales is not automatically the most profitable product, and sales may have been constrained by stockouts.

14. Category Revenue Analysis

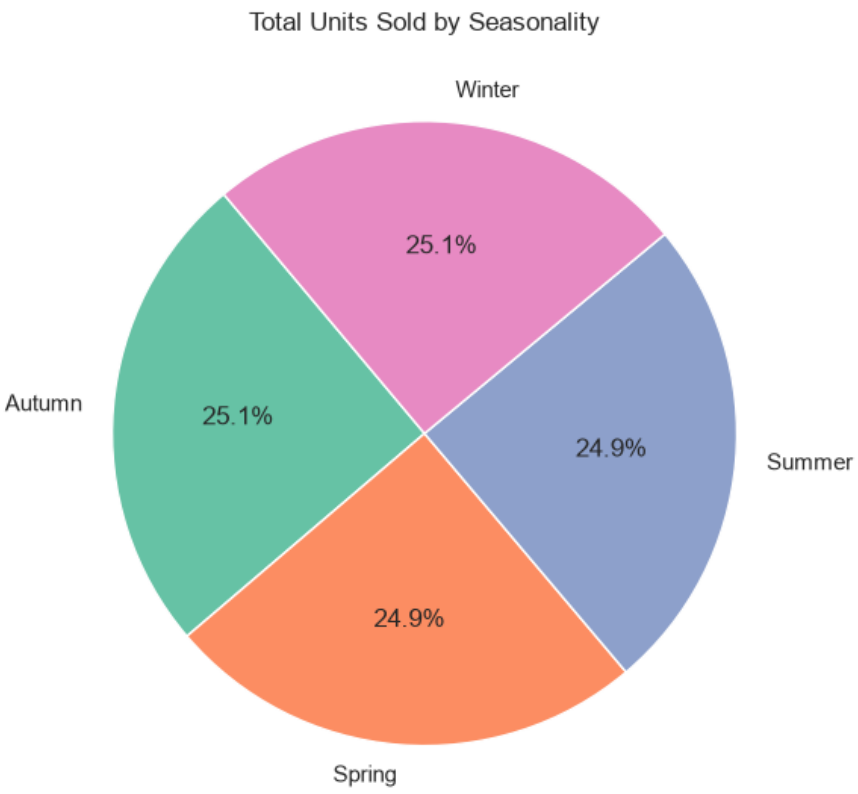


Category-level revenue comparison helps identify the categories contributing the largest recorded sales value. Management can use this view to investigate category performance and prioritize further analysis.

14.1 Numerical Summary

A category revenue table cannot be calculated from the selected CSV because a recognizable category field and revenue field were not both found. The existing chart, if present, is retained as a project output.

15. Seasonal Demand Distribution



The seasonal distribution chart helps examine how recorded unit demand varies across the seasonal groups represented in the project. Observed differences should be interpreted in the context of the dataset period.

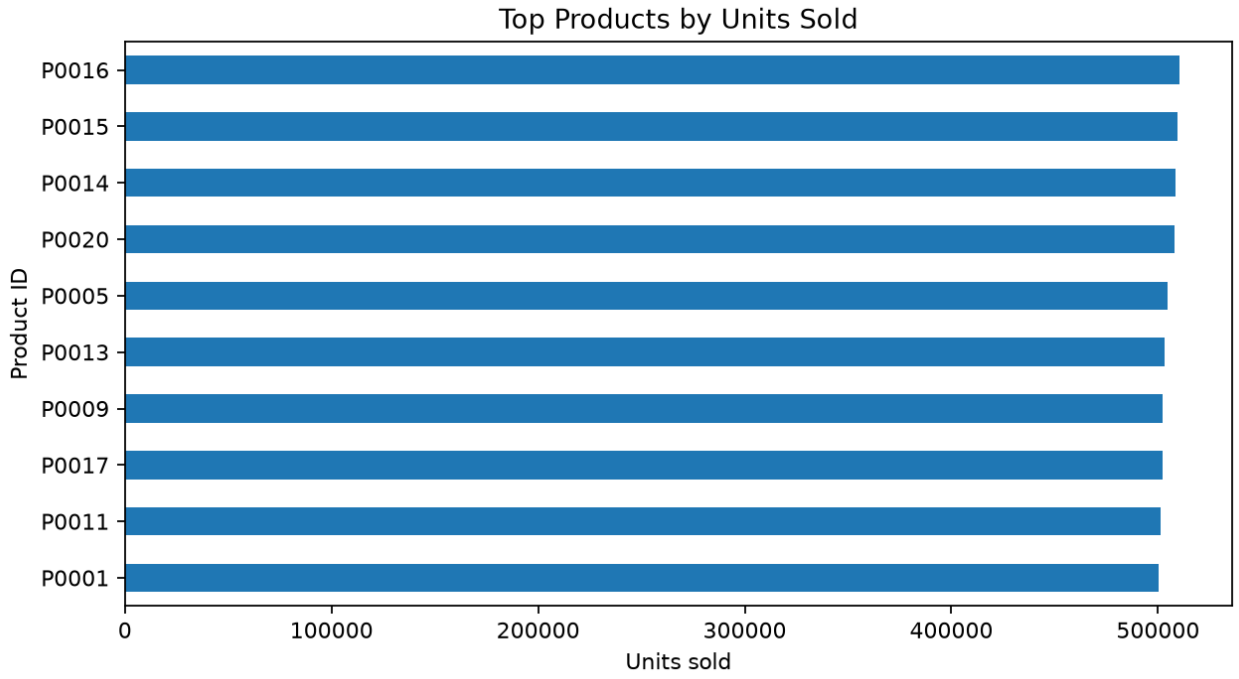
15.1 Understanding Seasonality

Seasonality refers to patterns that recur at regular intervals, such as weekly, monthly or yearly changes. In retail, demand may be affected by holidays, weather, promotions and shopping habits. A chart can suggest differences, but a reliable seasonal conclusion requires a suitable date range and consistent observations.

15.2 Business Application

If repeated seasonal demand is supported by the underlying data, inventory planning can be adjusted before expected high-demand periods. The size of any adjustment should be based on observed values and monitored after implementation.

16. Top-Demand Products



The top-demand products chart highlights products with relatively high recorded unit demand. These products may deserve closer monitoring when planning replenishment.

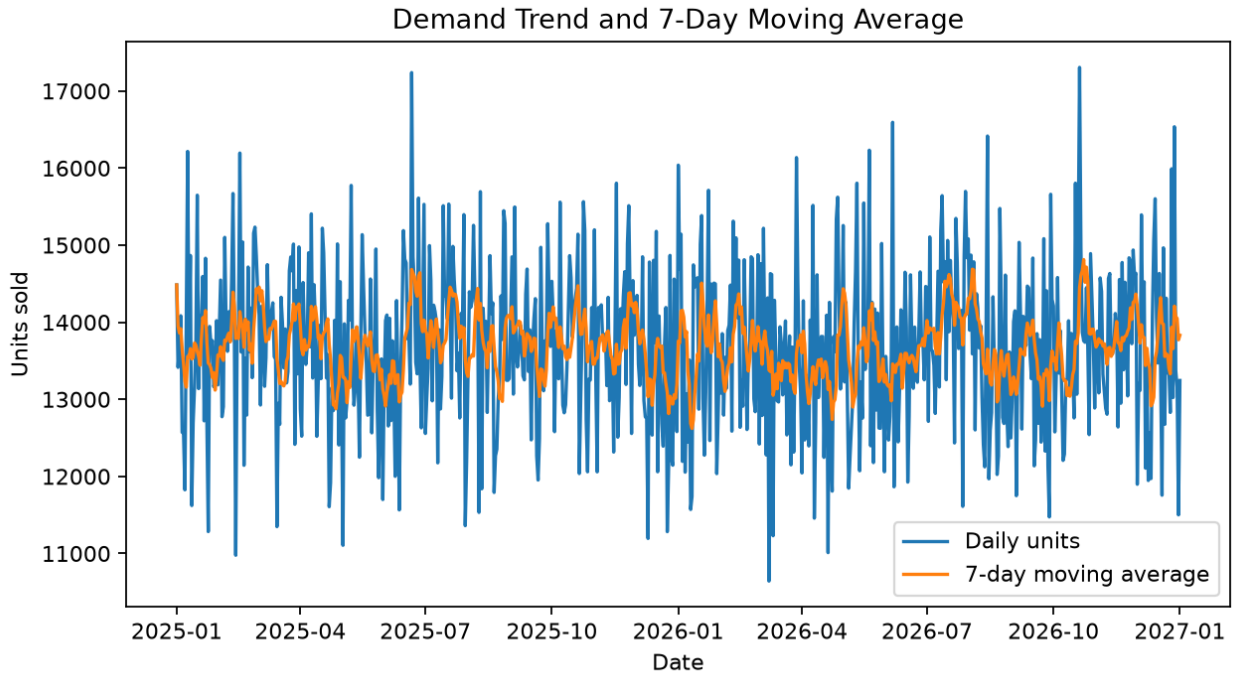
16.1 Product Demand Table

Product ID	Total Units Sold
P0016	511,891.00
P0015	511,377.00
P0014	510,122.00
P0020	509,998.00
P0005	506,490.00
P0013	504,228.00
P0009	503,376.00
P0017	503,372.00
P0011	502,581.00
P0001	501,049.00

16.2 Inventory Implications

Products with high recorded demand may require more frequent stock monitoring. However, replenishment decisions should also consider current stock, supplier lead time, minimum order quantities, storage cost and forecast uncertainty.

17. Demand Trend and Seven-Period Moving Average



The demand-trend chart compares demand with a seven-period moving average. Smoothing can make broad movements easier to observe, although it can also hide short-term changes.

17.1 Moving Average Concept

A moving average calculates the mean of a fixed number of recent observations. A seven-period moving average is calculated by taking the average of the current observation and the preceding six observations, when the data is ordered at the intended daily or weekly frequency.

7-period moving average = Mean of the latest 7 observations

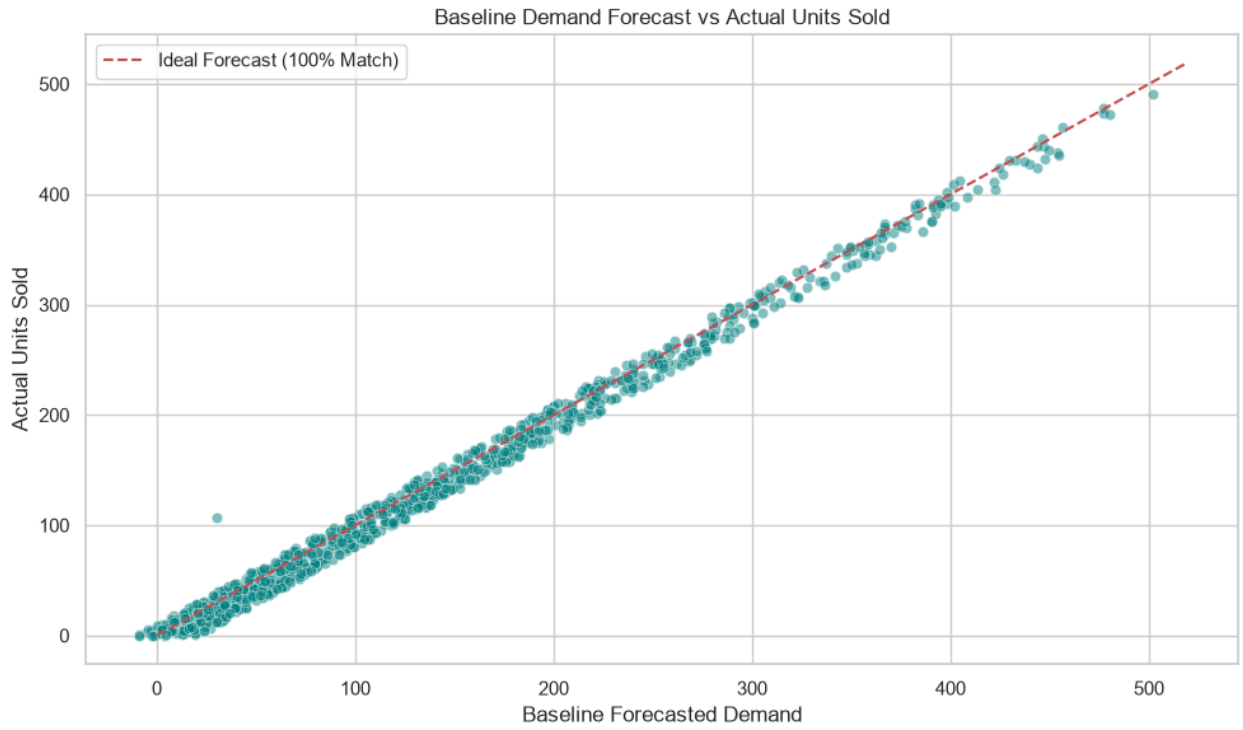
17.2 Benefits

- Reduces the visual effect of short-term fluctuations.
- Makes broader demand direction easier to inspect.
- Provides a simple baseline for forecasting.
- Can be compared with shorter or longer averaging windows.

17.3 Limitation

Moving averages respond slowly to sudden demand changes and may be misleading if records are not consistently spaced. The initial periods may also lack enough observations to calculate a complete window.

18. Forecast Versus Actual Demand



The forecast-versus-actual chart provides a visual comparison between forecast values and observed values. A close match suggests better alignment for the periods shown, but visual inspection alone is not a complete accuracy evaluation.

18.1 Baseline Forecast

A simple baseline forecast can use the mean demand of a recent window to estimate demand in a subsequent period. For example, the mean of the previous seven periods may be used as a baseline estimate for the next period.

Forecast for next period = Mean demand of the previous 7 periods

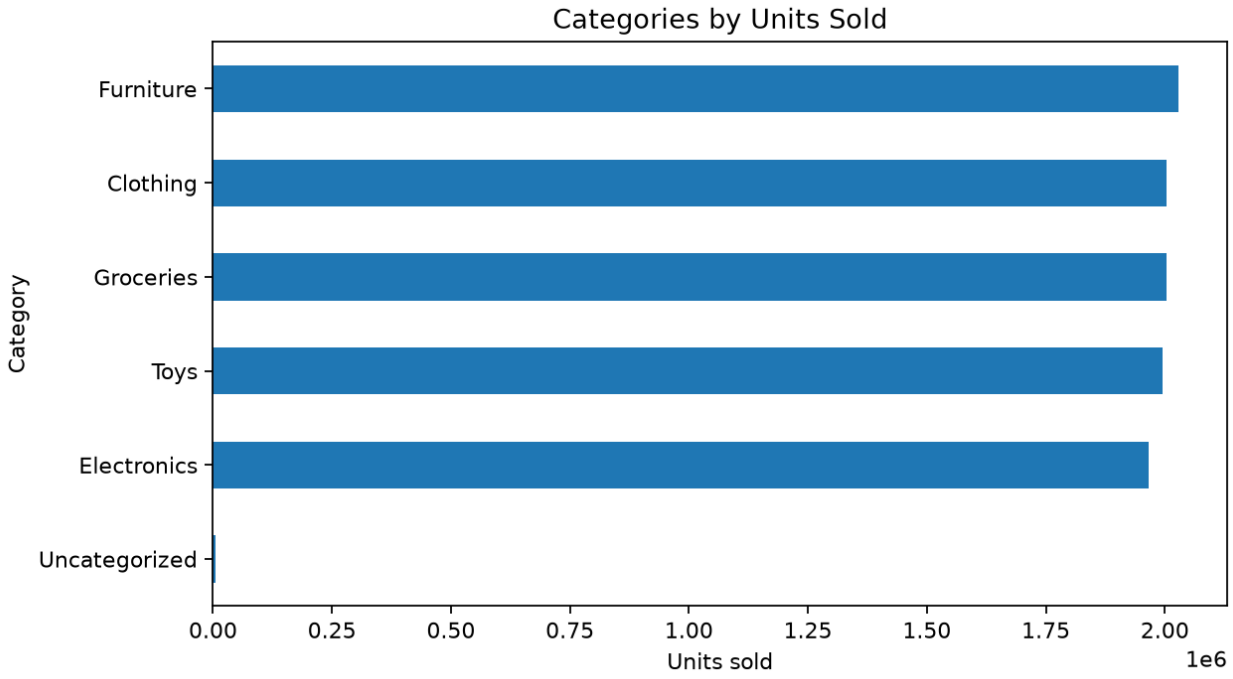
18.2 Forecast Evaluation

Forecast quality should be evaluated by comparing predictions with observations that were not used to produce those predictions. Suitable measures include Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

18.3 Interpretation

The existing chart is included as a visual comparison. This report does not claim a specific forecast accuracy percentage because that value should be calculated from the project's actual prediction and evaluation data.

19. High-Demand Categories



This chart compares categories identified as having higher demand in the existing analysis. The results can guide further product-level inventory review.

19.1 Category-Level Planning

Category-level analysis provides a broader view than looking at individual products. It can help determine where additional analysis or inventory monitoring may be useful. Category totals should be interpreted alongside the number of products in each category and the period represented.

19.2 Suggested Actions

- Review individual products within high-demand categories.
- Check whether high demand is consistent or caused by a short spike.
- Compare recent demand with available stock.
- Review supplier lead times before changing reorder quantities.
- Monitor category performance after replenishment changes.

20. Findings, Recommendations and Limitations

20.1 Findings Supported by the Report

- The selected dataset contains 73,501 records and 15 columns.
- The selected CSV currently contains 160 missing cells and 195 duplicate rows.
- The project includes visualizations for category revenue, seasonal demand, top products, moving averages, forecast comparisons and high-demand categories when those files are present.
- The data and chart outputs can support descriptive analysis, but forecasting conclusions require evaluation against observed outcomes.

20.2 Recommendations

- Monitor high-demand products and review their current stock.
- Use historical demand as one input for replenishment planning.
- Investigate missing values and duplicate records before analysis.
- Track forecast errors on periods excluded from model fitting.
- Compare multiple forecast windows before selecting a baseline.
- Review seasonal patterns over an adequate historical period.
- Keep the dataset, analysis scripts and report version aligned.

20.3 Limitations

- Historical demand does not guarantee future demand.
- Missing or inaccurate records may affect summaries.
- The selected dataset may not capture promotions, stockouts, supplier delays or external market conditions.
- A moving-average baseline may not capture complex trends.
- Existing charts may not expose all values needed for numerical forecast evaluation.

20.4 Ethical and Practical Considerations

Inventory recommendations should be reviewed by a responsible decision-maker before implementation. Stock decisions involve cost, customer service and operational constraints that may not be fully represented in the available dataset.

21. Conclusion, Future Scope and References

21.1 Conclusion

The Retail Inventory Analytics project demonstrates how Python-based data analysis can be used to examine retail demand and inventory-related patterns. The workflow covers dataset inspection, data-quality checks, descriptive summaries, visual analysis and interpretation of a simple forecasting approach.

The included charts provide different views of demand, category performance, time-based changes and forecast comparisons. Together, these outputs can help analysts identify questions that deserve further investigation. Practical inventory decisions should also account for stock availability, supplier lead times, costs and forecast errors.

21.2 Future Scope

- Add MAE, RMSE and other suitable forecast evaluation metrics.
- Compare seven-, fourteen- and thirty-period moving averages.
- Experiment with exponential smoothing or other forecasting models.
- Add explicit stockout, safety-stock and reorder-point analysis.
- Create an interactive dashboard for inventory monitoring.
- Automate scheduled dataset refresh and report generation.
- Evaluate forecasting performance separately for each product.

21.3 References

- Python documentation — https://docs.python.org/3/
- Pandas documentation — https://pandas.pydata.org/docs/
- NumPy documentation — https://numpy.org/doc/
- Matplotlib documentation — https://matplotlib.org/stable/
- ReportLab documentation — https://docs.reportlab.com/
- Project dataset, preprocessing scripts, analysis scripts and visualization files maintained in the project repository.

END OF REPORT